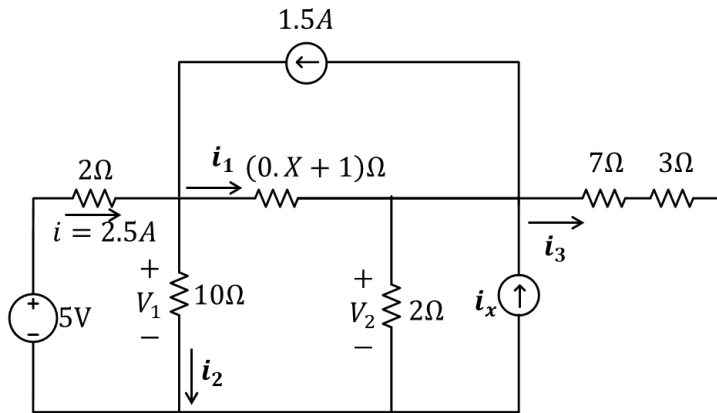
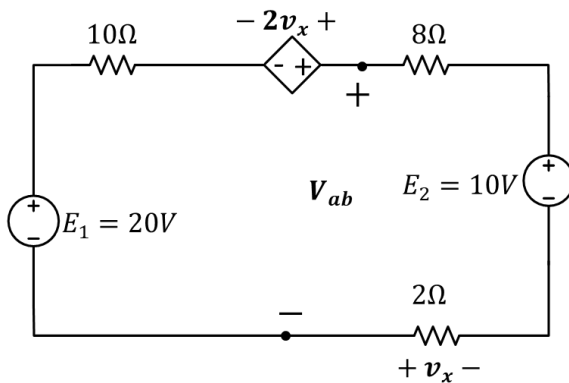


- **ANSWER THE FOLLOWING QUESTIONS**
- Figure in bracket [] next to each question indicates marks for that question
- Special Notation in the questions: **X**: Second last digit of your ID.

Question No. 1

- (a) Count the number of nodes in the circuit shown in **Fig.1**. Employing the basic circuit analysis concepts such as **Ohms Law, KCL, KVL etc.** determine i_1, i_2, V_1, V_2, i_3 and i_x for the circuit. Mention the corresponding concepts you have applied for determining each parameter while solving the circuit. [CO1-PO-a, P1, P3] [10]
- (b) For the circuit shown in **Fig.2**, determine V_{ba} and the power absorbed or delivered by the voltage sources. Which sources deliver power? Which sources absorb power? [10]

**Fig.1****Fig.2****Question No. 2**

- (a) Determine the equivalent resistance, R_{eq} , for the circuit shown in **Fig. 3**. Determine i , and i_2 using return and reduce method [15]

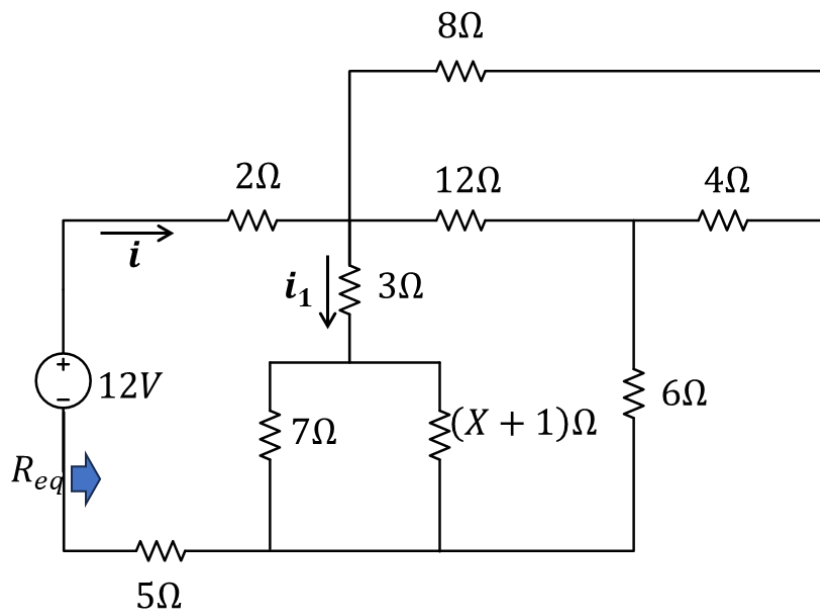


Fig.3.

- (b) Using Node analysis, determine the current flowing through the 2Ω and $(X + 6)\Omega$ resistor for the circuit shown in Fig.4. [10] [C02-PO-a, P1, P3]

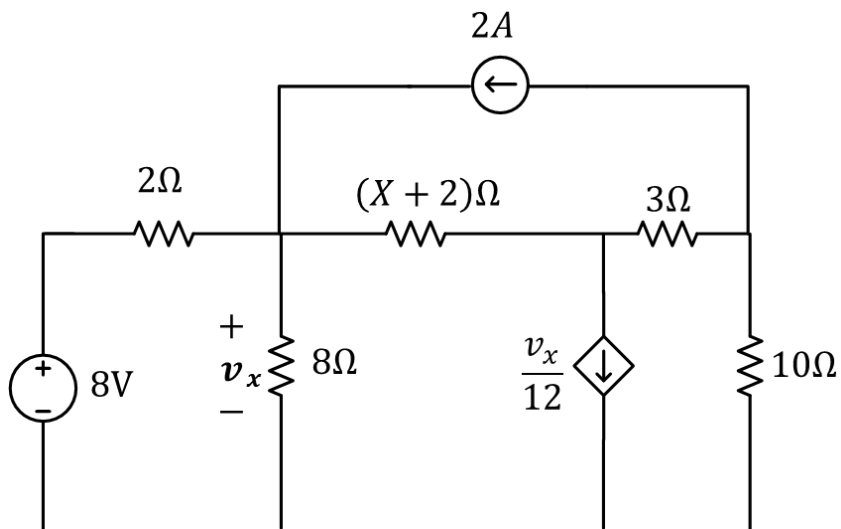


Fig.4

Name:

ID:

Bonus Question:

The charge flowing through a wire is plotted in **Fig.5**. Obtain the expressions of corresponding current $i(t)$ and sketch it for $10 < t < 20$ s. [8]

[Sketch the $i(t)$ on the graph provided below]

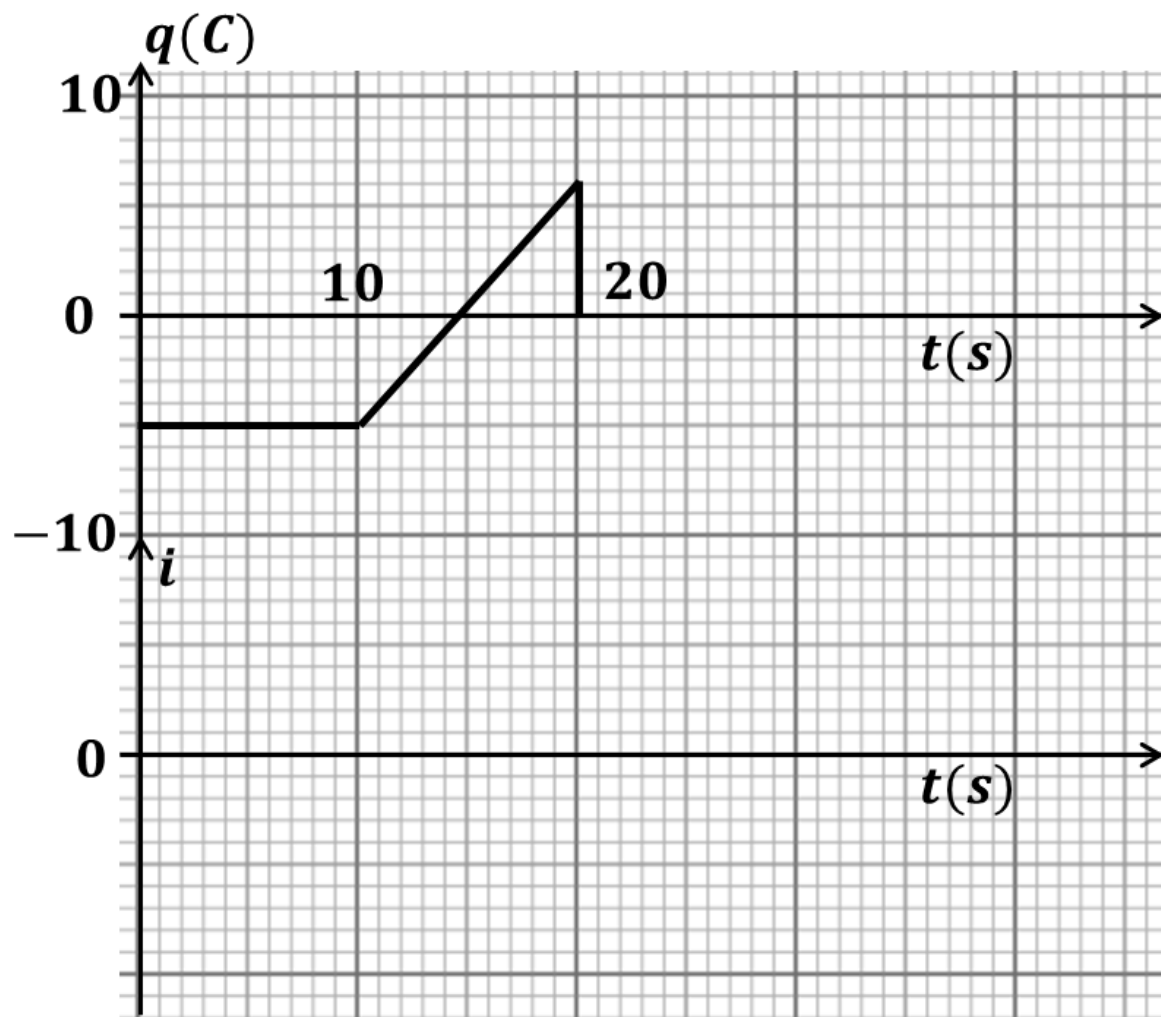


Fig.5.

THE END